

Project Executable Files

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction using AI/ML)
Maximum Marks	3 Marks

This section contains submission of all executable and deployable files for the project. All files are organized, named, and uploaded to the GitHub repository. The evaluator can run the solution locally using the provided files and instructions below.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	No

Step 2: File / Folder Structure

```

ML-0027-Human-Development-Index/
├── Dataset/
│   ├── Continent_HDI.csv
│   ├── countries.geojson
│   └── HDI.csv
├── Flask/
│   ├── templates/
│   │   ├── dashboard.html
│   │   ├── home.html
│   │   ├── indexnew.html
│   │   └── resultnew.html
│   └── app.py

```

```

|   └─ HDI.pkl
└─ Training/
    └─ HumDevIndex.ipynb
|
hdi-prediction-ml/ (GitHub repository)
|-- 1. Brainstorming & Ideation/
|-- 2. Requirement Analysis/
|-- 3. Project Design Phase/
|-- 4. Project Planning Phase/
|-- 5. Project Development Phase/
|-- 6. Project Testing/
|-- 7. Project Documentation/
|-- 8. Project Demonstration/
|-- README.md

```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not hosted — runs on local Flask development server (http://127.0.0.1:5000)
Login Credentials (Demo)	Not applicable — the application does not require user login or authentication
Platform / Hosting Provider	Local machine (Flask development server)
Repository Link	https://github.com/1904Spandhana/hdi-prediction-ml
Demo Video Link	To be added

Step 4: Run Instructions

Pre-requisites:

Python 3.x, Anaconda (recommended), VS Code or Jupyter Notebook. Required libraries: numpy, pandas, matplotlib, seaborn, scikit-learn, Flask.

Step 1: Clone the repository

```
git clone https://github.com/1904Spandhana/hdi-prediction-ml.git
```

Step 2: Install dependencies

```
pip install numpy pandas matplotlib scikit-learn flask seaborn
```

Step 3: Navigate to the Flask folder

```
cd ML-0027-Human-Development-Index/Flask
```

Step 4: Run the application

```
python app.py
```

Step 5: Open the application in a browser

Open <http://127.0.0.1:5000> in your browser

Step 6: Use the application

Click 'Start Prediction', enter Life Expectancy, Mean Years of Schooling, GNI per capita, and Internet Users, then click 'Predict HDI' to view the result. Visit /Dashboard to view the HDI dashboards.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Application requires HDI.pkl model file to be present in the Flask folder; missing file causes a startup error	Resolved — model file is committed to the repository under Flask/HDI.pkl
2	Application is not hosted on a public server and requires local setup to run	Workaround — clone repo and follow Run Instructions (Step 4) to run locally
3	Dashboard charts may take a moment to render when all three visualizations load together on slower systems	Minor — does not affect core prediction functionality